# Florida Hurricane Catastrophe Bond

## Three-layer per-occurrence structure for a hypothetical homeowners insurer

Teaching-grade illustration: simulated event set of hurricane-strength Florida landfalls fitted to HURDAT2 (1900–2019), hypothetical USD 100 billion portfolio, spread range from a regression on 56 public tranche prices.

**Illustrative only.** The sponsor and its portfolio are hypothetical. This document is not an offering of securities, a quote, a recommendation or investment advice. The loss model is a teaching-grade model built from public data, not a vendor catastrophe model (RMS, Verisk, CoreLogic); its expected losses are not comparable to those in a cat bond offering circular. CLIMADA was not used. Spread figures are regression outputs, not prices.

Generated from the results files of the cat-bond-issuance-model repository (scripts/make_pitchbook.py). Every number in this document is read from a file in `results/`.

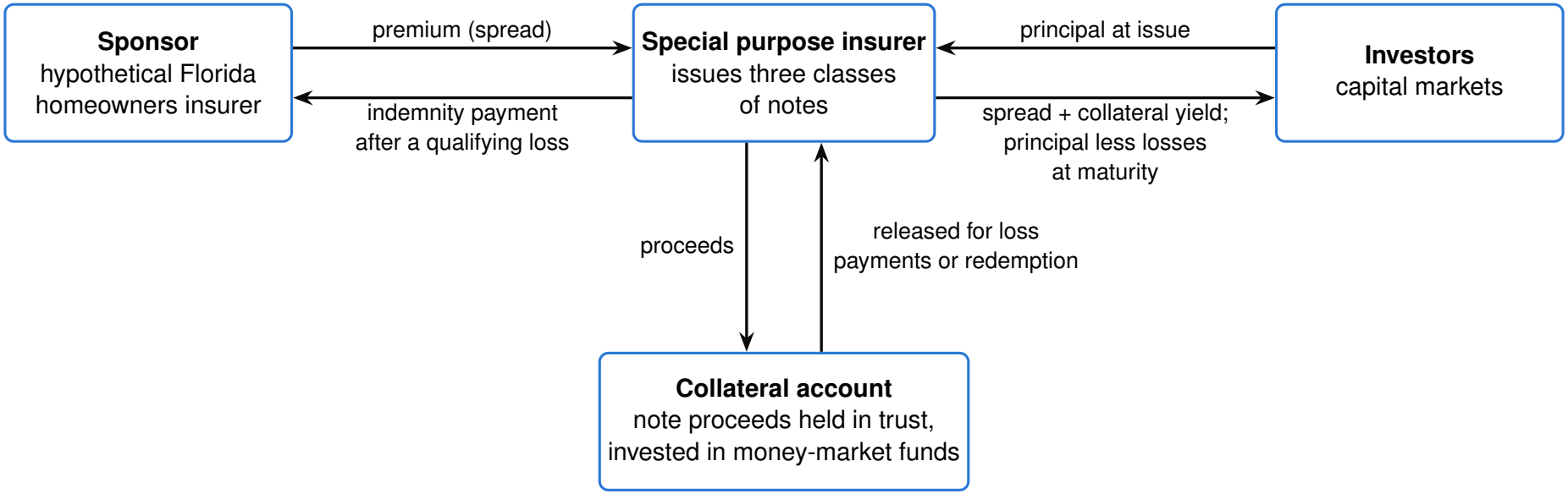
# Transaction summary (hypothetical)

Three adjacent per-occurrence layers of USD 500 million each (USD 1,500 million in total) on a hypothetical Florida homeowners portfolio, chosen by a rule fixed before any spread was fitted: the stack of grid layers whose expected losses are closest to 4%, 2% and 1% of limit.

| Layer | USD m | % TIV | AP % | EP % | EL % | EL band % | M1 spread % | M2 spread % |
|---|---|---|---|---|---|---|---|---|
| Layer 1 (bottom) | 500–1,000 | 0.50–1.00 | 5.67 | 2.70 | 3.90 | 2.19–5.34 | 11.5 [5.8, 22.5] | 12.1 [7.6, 19.2] |
| Layer 2 (middle) | 1,000–1,500 | 1.00–1.50 | 2.69 | 1.45 | 1.98 | 0.97–2.85 | 7.9 [4.1, 15.0] | 8.8 [5.7, 13.7] |
| Layer 3 (top) | 1,500–2,000 | 1.50–2.00 | 1.44 | 0.82 | 1.09 | 0.44–1.59 | 5.7 [3.0, 10.9] | 6.7 [4.2, 10.5] |

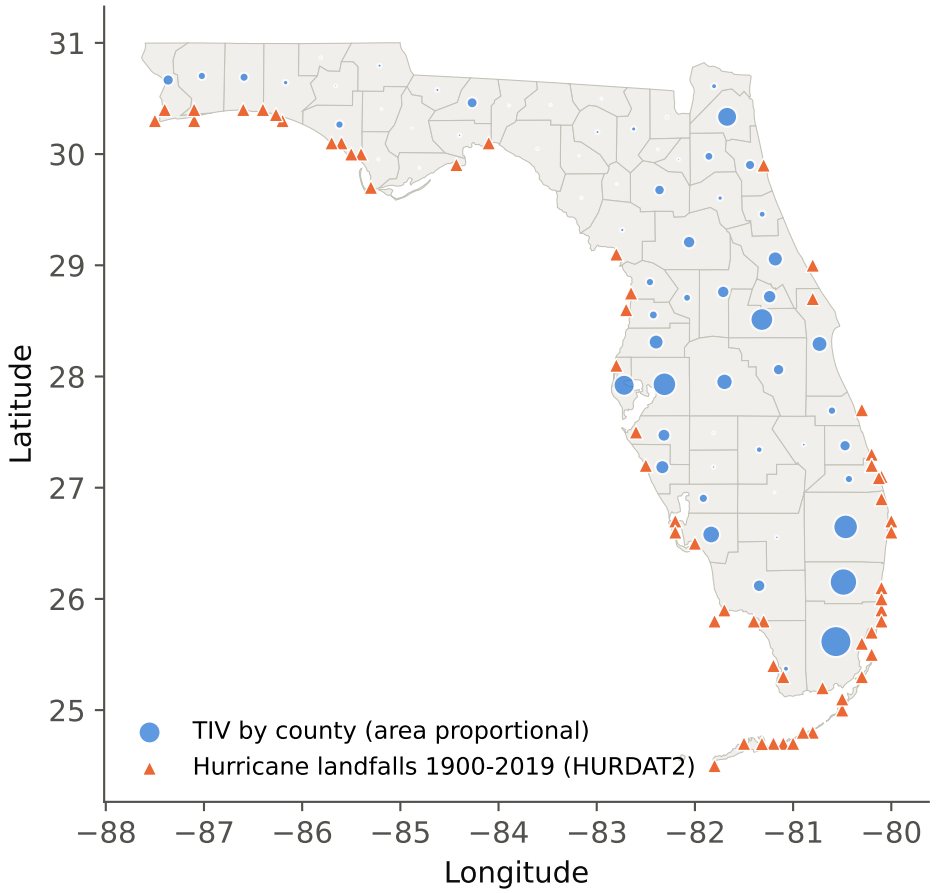
- AP: probability that at least one event in a year reaches the attachment. EP: probability that the whole layer is lost within a year. EL: mean annual principal loss over the limit (annual loss capped at the limit). All annual, from 100,000 simulated years (seed 1).
- EL band: 5% and 95% quantiles over 200 refits of the hazard model on bootstrap samples of the 1900–2019 years (frequency, intensity and landfall pool only, not model form).
- Spreads: median and 90% prediction interval from M1 (log spread on log EL) and M2 (plus issue-year effects, 2025) fitted on public layers. A regression output, not a quote.
- Probability of any principal loss over a 3-year term: 16.1%, 7.8% and 4.3% (bottom, middle, top).

# Structure




- Trigger: indemnity on the sponsor's (hypothetical) portfolio losses, per occurrence; every event in a year can erode principal, capped at each layer's limit.
- Layers attach at 0.50%, 1.00% and 1.50% of total insured value and exhaust at 2.00% for the top layer; they sit above the sponsor's retention.
- The generic structure is shown for teaching; legal terms, reset, extension and collateral rules of a real transaction are not modelled.

# Exposure summary (hypothetical portfolio)




Total insured value USD 100 billion (hypothetical) spread over the 67 Florida counties in proportion to 2020 Census households (UF BEBR table), placed at county centroids. The ten largest counties hold 59.8% of the value.

| County | Households 2020 | Share % | TIV USD m |
|---|---|---|---|
| Miami-Dade | 967,414 | 11.3 | 11,343 |
| Broward | 756,657 | 8.9 | 8,872 |
| Palm Beach | 607,880 | 7.1 | 7,127 |
| Hillsborough | 559,949 | 6.6 | 6,565 |
| Orange | 519,437 | 6.1 | 6,090 |
| Pinellas | 442,789 | 5.2 | 5,192 |
| Duval | 399,759 | 4.7 | 4,687 |
| Lee | 318,303 | 3.7 | 3,732 |
| Polk | 272,418 | 3.2 | 3,194 |
| Brevard | 257,026 | 3.0 | 3,014 |

No real insurer's book is used. A real portfolio differs by footprint, distance to the coast, construction and policy terms; deductible 2% of value, applied to the county mean damage ratio.

# Hazard and loss model; what was checked and what is weak

**Model.**

- Florida hurricane landfalls 1900–2019 (HURDAT2): 67 landfalls, mean 0.558 per year (95% 0.433–0.709). Poisson kept (negative binomial test p = 0.097); trend p = 0.23.
- Intensity: generalized Pareto on wind above 64 kt (shape -0.359, scale 39.3 kt, KS distance 0.086).
- Track: historical landfall position, heading and speed resampled with jitter; Holland wind profile; Kaplan–DeMaria inland decay; Emanuel (2011) damage function.
- Landfall definition (1966–1984): on 6-hourly fixes only, the geometric rule finds 58 of 62 flagged landfalls of other years, with winds 4.0 kt lower on average.

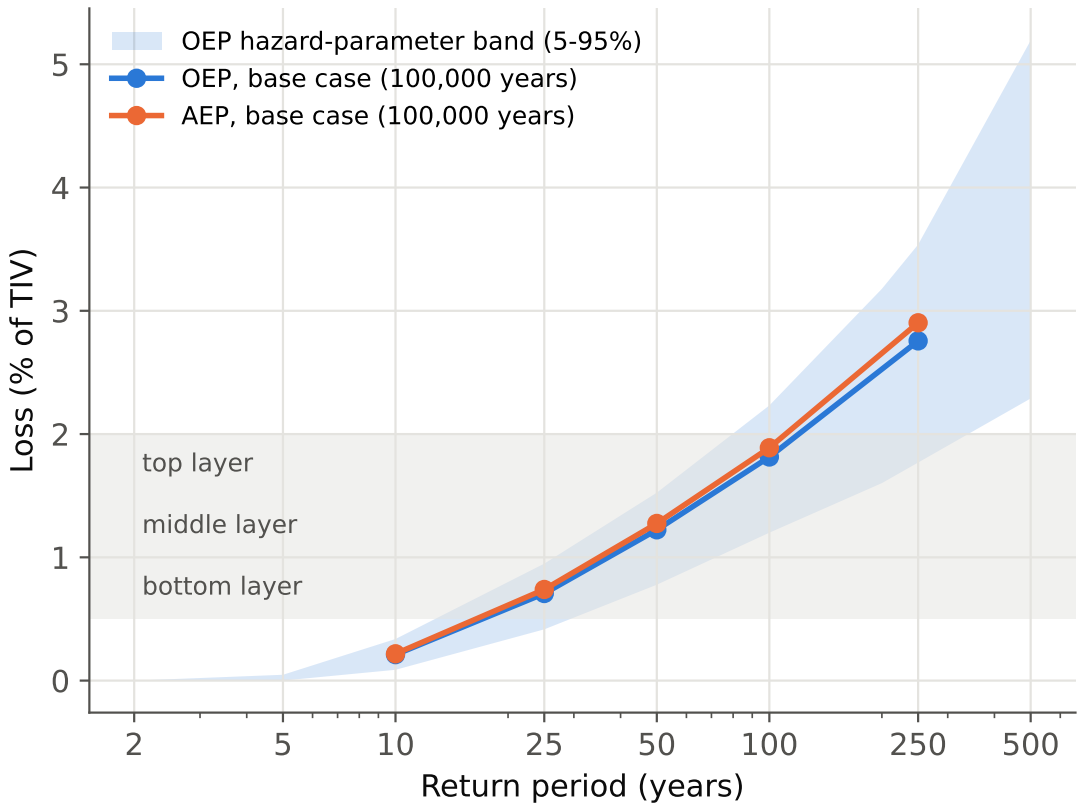
**Weak points.**

- Fit 1900–1979, predict 1980–2019: 25.0 hurricanes predicted, 17 observed (p = 0.21). No category falls outside its interval, but the intervals are wide: the check has little power and says nothing about losses.
- Window 1950–2019 gives 0.457 per year and the negative binomial model (p = 0.013); the layer ELs fall to 1.04% for the middle layer (sensitivity page).
- No surge, rain, terrain or building inventory; straight tracks; losses not calibrated to any insured loss.

**Out-of-sample check by Saffir-Simpson category** (counts 1980–2019)

| Category | Predicted | Bootstrap 90% PI | Observed | p (two-sided) |
|---|---|---|---|---|
| Cat 1 | 10.8 | 4–17 | 6 | 0.35 |
| Cat 2 | 5.2 | 2–10 | 3 | 0.51 |
| Cat 3 | 4.6 | 1–10 | 4 | 0.86 |
| Cat 4 | 3.3 | 0–7 | 2 | 0.71 |
| Cat 5 | 1.0 | 0–3 | 2 | 0.47 |
| Cat 3-5 (major) | 9.0 | 4–16 | 8 | 0.78 |
| All hurricanes | 25.0 | 15–36 | 17 | 0.21 |

# Loss exceedance curve with hazard-parameter band




| Return period | OEP % TIV | SE | 5–95% band | AEP % TIV |
|---|---|---|---|---|
| 10 | 0.21 | 0.004 | 0.09–0.34 | 0.22 |
| 25 | 0.71 | 0.009 | 0.42–0.95 | 0.74 |
| 50 | 1.22 | 0.019 | 0.78–1.52 | 1.27 |
| 100 | 1.81 | 0.027 | 1.20–2.23 | 1.89 |
| 250 | 2.76 | 0.063 | 1.77–3.54 | 2.90 |

1-in-100 occurrence loss 1.81% of TIV (Monte Carlo SE 0.027); across 200 hazard refits the 5–95% range is 1.20–2.23%. Average annual loss 0.101% of TIV.

The band reflects uncertainty in frequency, intensity and the landfall pool (10,000 years per refit). Model-form choices move losses much more (sensitivity page). Shaded horizontal bands: the three layers.

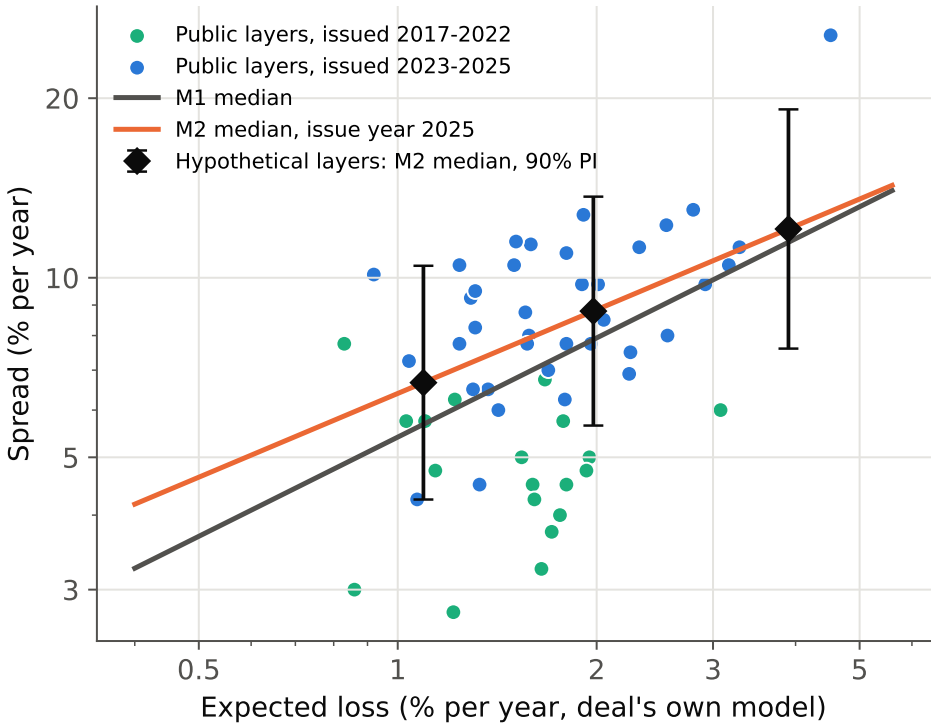
# Layer analytics

**Candidate layers with a limit of 0.5% of TIV** (base case, 100,000 years)

| Attach % TIV | Exhaust % TIV | AP % | EP % | EL % | MC SE | P(loss in 3 y) % |
|---|---|---|---|---|---|---|
| 0.25 | 0.75 | 9.14 | 3.81 | 5.93 | 0.069 | 25.1 |
| 0.50 | 1.00 | 5.67 | 2.70 | 3.90 | 0.057 | 16.1 |
| 0.75 | 1.25 | 3.76 | 1.96 | 2.74 | 0.049 | 10.9 |
| 1.00 | 1.50 | 2.69 | 1.45 | 1.98 | 0.042 | 7.8 |
| 1.25 | 1.75 | 1.95 | 1.08 | 1.46 | 0.036 | 5.7 |
| 1.50 | 2.00 | 1.44 | 0.82 | 1.09 | 0.031 | 4.3 |
| 1.75 | 2.25 | 1.08 | 0.64 | 0.83 | 0.027 | 3.2 |
| 2.00 | 2.50 | 0.82 | 0.51 | 0.65 | 0.024 | 2.4 |
| 2.25 | 2.75 | 0.63 | 0.40 | 0.51 | 0.022 | 1.9 |
| 2.50 | 3.00 | 0.51 | 0.33 | 0.41 | 0.019 | 1.5 |
| 2.75 | 3.25 | 0.40 | 0.28 | 0.33 | 0.018 | 1.2 |
| 3.00 | 3.50 | 0.33 | 0.23 | 0.28 | 0.016 | 1.0 |

- Each row is a per-occurrence layer; EL falls quickly with the attachment, AP faster than EP.
- The chosen layers are the rows attaching at 0.50, 1.00 and 1.50% of TIV, with EL 3.90, 1.98 and 1.09% (Monte Carlo SE 0.06, 0.04, 0.03).
- EP counts total loss of a layer within a year, including by several partial events.
- The reference layer (1.20–1.80% of TIV, the rounded 1-in-50 to 1-in-100 occurrence losses) has EL 1.47%.

# Pricing: regression on public tranche prices




56 distinct layers, 37 series, 14 sponsors, issued 2017–2025 (Artemis pages). Deal ELs come from each deal's own vendor model.

| Model | Interval method | b | 95% interval | p (b = 1) |
|---|---|---|---|---|
| M1 | HC3, t(n-k) | 0.553 | [0.179, 0.926] | 0.020 |
| M1 | CR1 series, t(G-1) | 0.553 | [0.174, 0.931] | 0.022 |
| M1 | wild cluster bootstrap series | 0.553 | [0.094, 0.966] | 0.036 |
| M1 | CR1 sponsor, t(G-1) | 0.553 | [0.091, 1.014] | 0.056 |
| M1 | wild cluster bootstrap sponsor | 0.553 | [0.167, 1.155] | 0.157 |
| M2 | HC3, t(n-k) | 0.466 | [0.128, 0.805] | 0.003 |
| M2 | CR1 series, t(G-1) | 0.466 | [0.156, 0.776] | 0.001 |
| M2 | wild cluster bootstrap series | 0.466 | [0.115, 0.830] | 0.008 |
| M2 | CR1 sponsor, t(G-1) | 0.466 | [0.124, 0.808] | 0.005 |
| M2 | wild cluster bootstrap sponsor | 0.466 | [0.146, 0.871] | 0.016 |
| M3 | HC3, t(n-k) | 0.492 | [0.141, 0.844] | 0.006 |
| M3 | CR1 series, t(G-1) | 0.492 | [0.174, 0.810] | 0.003 |
| M3 | CR1 sponsor, t(G-1) | 0.492 | [0.127, 0.858] | 0.010 |

Pre-specified rule: proportionality (b = 1) is rejected only if the series wild bootstrap interval (9,999 draws) excludes 1 in M1 and M2. Main sample: M1 0.553 [0.094, 0.966], M2 0.466 [0.115, 0.830]: rejected, but in only 2 of 5 pre-specified samples (next page). R-squared 0.21 (M1), 0.69 (M2): issue year explains more than EL. Holding out a series, M2 misses by 3.02 points of spread (RMSE).

# Pricing: robustness and comparable deals

### Slope in every pre-specified sample

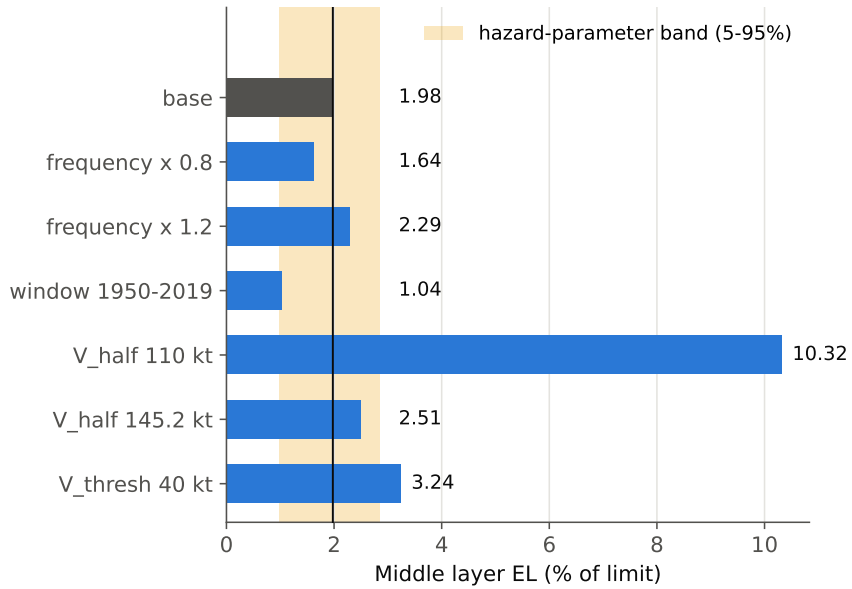
| Sample | M1 b [wild, series] | M2 b [wild, series] | b = 1 rejected |
|---|---|---|---|
| main | 0.553 [0.094, 0.966] | 0.466 [0.115, 0.830] | yes |
| w/o guidance prices | 0.619 [0.169, 1.016] | 0.491 [0.112, 0.896] | no |
| w/o discount notes | 0.635 [0.174, 1.033] | 0.510 [0.163, 0.877] | no |
| w/o launch-plan EL | 0.556 [0.094, 0.972] | 0.510 [0.171, 0.881] | yes |
| w/o all flagged rows | 0.703 [0.262, 1.084] | 0.547 [0.160, 0.944] | no |

**Caveats.** Programs were chosen by name; two sponsors give most layers; median spread/EL multiple by issue year runs from 2.36 (2018) to 6.67 (2024); ELs are offering-stage figures from different vendor models; three prices are guidance and four are discount-note equivalents. Leave-one-sponsor-out RMSE 3.20 points. The model's EL and the deals' ELs are on different scales.

### Five nearest public layers by EL and year

| Layer | Deal | Class | Year | EL % | Spread % | Multiple |
|---|---|---|---|---|---|---|
| bottom | Integrity Re III (Series 2025-1) | D | 2025 | 4.52 | 25.50 | 5.64 |
| bottom | Gateway Re (Series 2025-1) | A | 2025 | 3.17 | 10.50 | 3.31 |
| bottom | Nature Coast Re (Series 2025-1) | A | 2025 | 2.92 | 9.75 | 3.34 |
| bottom | Alamo Re (Series 2024-1) | C | 2024 | 3.29 | 11.25 | 3.42 |
| bottom | Everglades Re II (Series 2025-1) | D | 2025 | 2.80 | 13.00 | 4.64 |
| middle | Integrity Re III (Series 2025-1) | B1 | 2025 | 2.01 | 9.75 | 4.85 |
| middle | Everglades Re II (Series 2025-1) | B | 2025 | 1.90 | 9.75 | 5.13 |
| middle | Palm Re (Series 2025-1) | A | 2025 | 1.80 | 7.75 | 4.31 |
| middle | Gateway Re (Series 2025-1) | AA | 2025 | 1.79 | 6.25 | 3.49 |
| middle | Cape Lookout Re (Series 2025-1) | A | 2025 | 2.24 | 6.90 | 3.08 |
| top | Gateway Re (Series 2025-1) | AAA | 2025 | 1.07 | 4.25 | 3.97 |
| top | Purple Re (Series 2025-1) | A | 2025 | 1.04 | 7.25 | 6.97 |
| top | Purple Re (Series 2025-1) | B | 2025 | 1.24 | 7.75 | 6.25 |
| top | First Coast Re IV (Series 2025-1) | A | 2025 | 1.30 | 6.50 | 5.00 |
| top | Gateway Re (Series 2025-1) | C1 | 2025 | 1.31 | 8.25 | 6.30 |

# Sensitivities of layer EL and implied spread




| Variant (* in protocol) | EL bottom | EL middle | EL top | M2 spread middle % |
|---|---|---|---|---|
| base * | 3.90 | 1.98 | 1.09 | 8.79 |
| frequency x 0.8 * | 3.15 | 1.64 | 0.91 | 8.05 |
| frequency x 1.2 * | 4.57 | 2.29 | 1.25 | 9.42 |
| intensity: KDE instead of GPD | 3.85 | 1.88 | 1.07 | 8.58 |
| window 1950-2019 * | 2.33 | 1.04 | 0.43 | 6.51 |
| window 1900-2024 | 4.00 | 2.03 | 1.08 | 8.90 |
| V_half 110 kt * | 13.98 | 10.32 | 7.95 | 19.00 |
| V_half 145.2 kt * | 4.67 | 2.51 | 1.44 | 9.82 |
| V_thresh 40 kt * | 6.11 | 3.24 | 1.83 | 11.06 |
| deductible 0% | 6.08 | 3.11 | 1.72 | 10.86 |
| deductible 5% | 2.37 | 1.18 | 0.65 | 6.90 |
| RMW x 0.8 | 2.15 | 1.05 | 0.60 | 6.53 |
| RMW x 1.2 | 5.92 | 3.35 | 2.08 | 11.24 |
| no inland decay | 12.41 | 9.66 | 7.83 | 18.42 |

Same seed and 100,000 years per variant. Variants that change only loss settings reuse the base case's events, the KDE variant keeps their counts, positions and tracks with other winds, and the frequency and window variants are independent runs. The half-damage wind of the damage function moves the middle-layer EL to 10.32% and the 1950–2019 window to 1.04%; the public deals span EL 0.83–4.52%, so spreads mapped from ELs outside that range are extrapolations.

## Limitations

- **Teaching-grade loss model.** County-centroid exposure, a parametric wind field, straight tracks, no storm surge, rainfall, terrain, building inventory or demand surge; no calibration to insured losses. Model-form choices (damage function, decay, wind radius) move layer EL far more than the hazard-parameter band.
- **Hypothetical portfolio.** Value spread by households, not insured values; one deductible applied to county mean damage.
- **Hazard record.** 120 years of landfalls. Only the 67 hurricane-strength landfalls of the definition are counted: tropical storms are excluded by design, and 7 hurricanes that came within 50 km of the Florida coast without being counted (Ivan 2004 and Matthew 2016 among them) add nothing, so simulated losses are biased low in a way the band does not show. The 1966–1984 landfall flags are missing in the data file and a geometric rule replaces them; on 6-hourly fixes the rule finds 58 of 62 landfalls with winds 4.0 kt lower on average. Intensity is drawn independently of location.
- **Deal data.** Only Artemis pages, programs chosen by name, offering-stage ELs from different vendor models, a few guidance and discount-note prices; transcription re-checked on 30 rows, not the source itself.
- **Pricing regression.** Only 21 of the 56 layers list Florida alone, 19 list no Florida and 15 cover other perils too, and the regression has no region or peril term. Year effects absorb the market cycle; errors correlated within series and sponsor; leave-one-series-out RMSE 3.02 points of spread; mapping this model's EL through it assumes a common EL scale that is not shown.
- **Not modelled.** Collateral yield, term structure, reset and extension provisions, ratings, investor demand, fees and expenses.

# Appendix: assumptions and sources

| Item | Assumption or source |
|---|---|
| Track data | HURDAT2 Atlantic best tracks (NOAA NHC; Landsea and Franklin 2013), GitHub copies pinned by SHA-256 |
| Calibration window | 1900–2019 (pre-specified); 1950–2019 and 1900–2024 as sensitivities |
| Wind field | Holland (1980) profile; radius of maximum wind and B from Willoughby and Rahn (2004) |
| Inland decay | Kaplan and DeMaria (1995), NHC operational constants |
| Damage | Emanuel (2011): threshold 50 kt, half-damage 150 kt (110 kt and 145.2 kt as sensitivities) |
| Exposure | 2020 Census households by county (UF BEBR); TIV hypothetical; 2% deductible |
| Simulation | 100,000 years, seed 1; hazard-parameter band from 200 year-bootstrap refits |
| Deal terms | Artemis deal directory pages, accessed 2026-11-05; two transcription checks (the first by the collector) |
| Regression | OLS of log spread on log EL; HC3 (MacKinnon and White 1985), cluster-robust (Cameron and Miller 2015), wild cluster bootstrap (Cameron, Gelbach and Miller 2008; Roodman et al. 2019) |
| Protocol | All models, samples and the layer rule fixed in docs/pricing_protocol.md before the first fit |
| Files | results/transaction/, results/pricing/, results/eventset/, results/param_uncertainty/ |

**References.** Cameron, Gelbach and Miller (2008), Review of Economics and Statistics 90(3); Cameron and Miller (2015), Journal of Human Resources 50(2); Emanuel (2011), Weather, Climate, and Society 3; Holland (1980), Monthly Weather Review 108(8); Kaplan and DeMaria (1995), Journal of Applied Meteorology 34; Landsea and Franklin (2013), Monthly Weather Review 141; MacKinnon and White (1985), Journal of Econometrics 29(3); Roodman, MacKinnon, Nielsen and Webb (2019), Stata Journal 19(1); Willoughby and Rahn (2004), Monthly Weather Review 132.